Learning Fast Dynamic Imitation Online

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https://johnyao93.github.io/spar

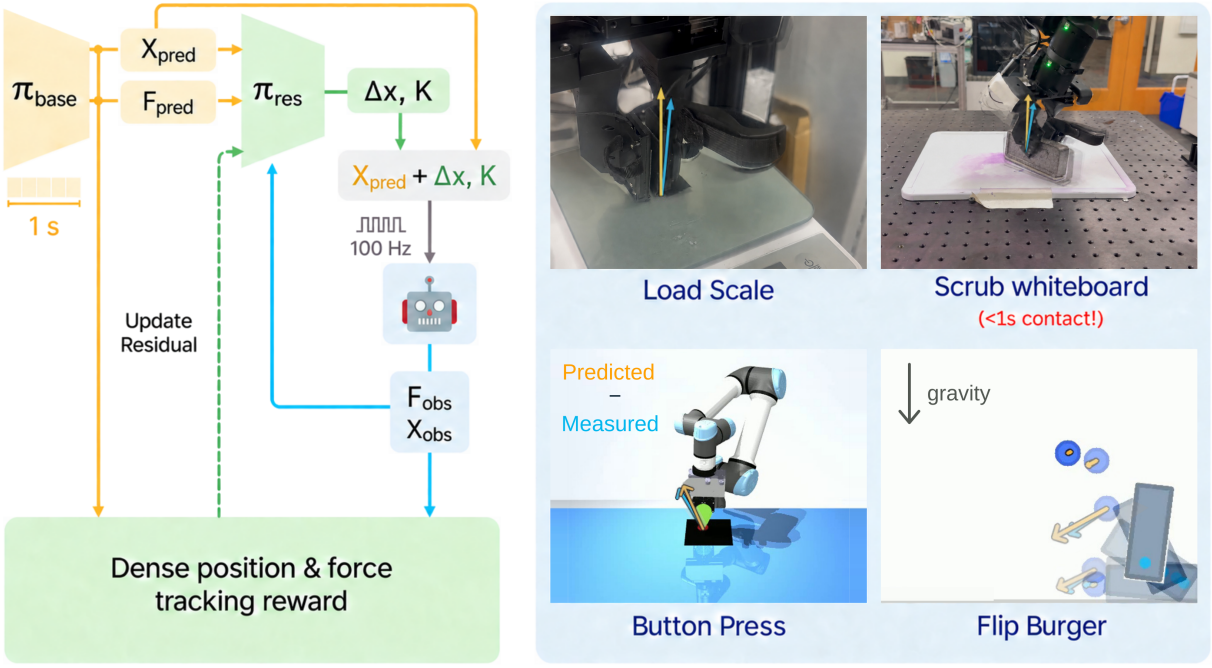


Fig. 1: **Overview of SPAR.** We present a practical real-world RL approach to train a *learned controller* to realize forceful demonstrations from only observed positions. SPAR trains a residual policy that outputs position deltas and controller stiffnesses based on a simple dense reward for matching *forces* predicted from a base policy. We find SPAR achieves success rates up to 90% above position-only policies across a wide variety of embodiments in less than twenty minutes of wall-clock time.

***Abstract*— Recent advances in imitation learning in robotic manipulation include optimized data collection interfaces offering scalability and speed by fixing handles near the end-effector. While such designs are ideal for capturing natural human movement, treating observed positions as actions often leads to task failure when precise force application or fast motion is necessary. Proposed analytical methods combined with force sensors attached at the end effector partially recover vital action information. However, these methods rely on specific assumptions about the environment, often violated by high-speed movement, and require tuning contact models for task success. We propose using predicted force as a reliable signal to discover actions through real-world interaction via residual reinforcement learning (RL), replicating the predicted forces. With SPAR, we efficiently learn fast and compliant robot interaction from position-only data to solve dynamic and force-sensitive tasks such as flipping a book, loading a scale, and scrubbing a whiteboard at speed, reaching over 80% success rate on all in under twenty minutes of real-world RL.**

I. INTRODUCTION

Data collection techniques that leverage kinesthetic teaching [1]–[8] and hand-held devices [9]–[11] promise significant benefits over robot teleoperation: directly moving a robot or handheld device provides operators with direct force feedback, leading to faster, higher-quality motion [1], while improved portability can enable more scalable data collection [9]. However, the recorded data only includes position trajectories while executing a task and does not include the motor actions required to realize those positions to ultimately accomplish the task. Moreover, it may not be feasible to accurately track the demonstrated motions at the demonstrated speed due to finite stiffness in robot actuators, often necessitating slower playback. In other cases, even exact motion tracking may fail for dynamic, force-sensitive manipulation tasks, when forces must also

be mimicked closely. In this paper, we aim to learn from kinesthetic data with both motions and forces to accurately recover actions and ultimately complete challenging force-sensitive manipulation tasks.

We use the term *dynamic imitation* to refer to the problem of matching both motion and applied forces, particularly using data that lacks action information. Prior works have studied analytical methods for this problem, calculating virtual targets with adaptive stiffness in order to replicate forces recorded in a trajectory [12]–[15]. Unfortunately, these methods rely on carefully calibrated force readings and require extensive manual gain tuning, which can vary per task. We aim to develop an approach that applies to noisy, uncalibrated force sensors and automatically adapts to different tasks.

Given force prediction reliability [16], [17] and the absence of actuator commands in haptic data collection, utilizing end-effector force sensors remains effective for capturing force information during demonstration. Analytical methods use these forces to create action information in the dataset for downstream imitation learning. However, the restrictive assumption of analytical approaches remains difficult to bridge to many real-world challenges. For example, most models assume slow motion to isolate measured forces to contact forces, restricting application. Therefore, we need an approach that can discover action correction while remaining reactive at high speeds to imitate the contact dynamics of natural human manipulation. Further, given high accuracy of the position reference and stiff gains, only small corrections are necessary to reproduce expected forces.

To this end, we present SPAR, which learns dynamic imitation online with real-world residual RL. Sitting as a learned controller underneath a base policy, the action space comprises residual actions and stiffness to adapt towards a simple and general reward, based only on the delta between predicted and measured position and force. To demonstrate its capability of adapting to high speed and natural human movement, we adapt the base policy trained using position and end-effector force data from ALOHA Lightning [1] to successfully accomplish forceful tasks in the real world such as the blind loading of a scale to a specific weight and scrubbing a large whiteboard mark. In total, we raise success rates on high-speed, forceful tasks from zero to over 80% using only position and force information.

Our main contribution is a method that:

1) Uses a simple and general reward capable of recovering actions that successfully track force with minimal position deviation.
2) Can learn to map complex and high-frequency contact predictions to residual actions conditioned by stiffness.
3) Is general and applies to any action space and any force sensor, even uncalibrated.
4) Is highly practical, learning contact-rich adjustments in less than twenty minutes of robot time.

II. Related Work

A. Learning From Observations

A broad class of works learn to recover actions from observation-only data, ranging from learning inverse dynamics models to recover actions from observed and future observations [18]–[20] to learning the actions required to achieve reached poses in human videos [19], [21]–[23] to direct motion reconstruction [24]. In our setting, we specifically concentrate on the instance where measured poses are recorded from robot manipulation trajectories, but not the controller actions required to reach them.

B. Analytical Methods

For precise force regulation, classical methods introduce compliance in interactions through different control paradigms such as impedance control [25], admittance control [26], and hybrid force-position control [27]. However, as such methods alone must be tuned to specific task settings, elements of these analytical approaches have been incorporated in robot learning methods for greater generality of application [14], [17], [28]. Typically, these hybrid works rely on similar hardware setups to ours where a force sensor is equipped on the end effector and interaction forces are recorded and trained upon, either in the form of directly predicting forces [16], [17], stiffness matrices [14], [29], or hybrid force-motion primitives [17], [28]. While effective in particular settings, such approaches still inherit many of the restrictive assumptions on the environment from analytical techniques. For example, many works rely on accurate force predictions which might either be sourced from an expensive Force/Torque sensor mounted on the wrist [14], or adapted to through a calibration technique on a powerful sensor [15]. In contrast, our method is able to use uncalibrated force readings, assuming consistency between test and training time. Other approaches assume slow motion when relating measured forces to contact dynamics [14], [30]. For example, [30] forgoes acceleration and velocity terms in their controller, while our policy learns residuals from tracking errors without such assumptions.

C. Reinforcement Learning for Compliance Control

Several methods learn compliant behavior through interaction via policy search [31]–[33]. [31] explicitly penalizes high gains when learning impedance, whereas our policy learns to soften through position and force tracking rewards alone. As real-world exploration can be costly, other approaches improve data efficiency with model-based learning of a hybrid force-position controller [30], or by training variable-impedance policies

in simulation before transferring to hardware [34]–[36]. [33] uses model-free RL to learn pose corrections and force-controller parameters for assembly, with rewards for goal attainment and reducing contact forces. In a similar vein, [37] learns residual corrections using tactile input, with a binary success reward. In contrast, our method learns to track *predicted* force profiles on a real robot directly, collecting realistic contacts for minimal train and test time gap. Closer to our setting, several works also initialize policies from demonstrations with live corrections in the real world, either through residual corrections from additional human feedback [38] or by augmenting demonstrations with approximate stiffness labels and fine-tuning the resulting policy using task-success and safety rewards [39]. Learning DAgger corrections can be efficient, but relies on human intervention which can be difficult to administer at high speed. In contrast to [39], SPAR requires neither hand labeled stiffnesses nor task reward.

D. Residual and Hierarchical Policies

Residual policies [40], [41] are a common approach for training high-frequency reactive policies. This typically involves a slower base policy, and a smaller residual policy that runs inference more frequently and predicts residual actions on top of the base policy's predictions. As residual policies may inference more frequently, they may react to faster streaming modalities than images, such as tactile or force information [42]–[44]. Our method specifically recovers residual actions needed to accomplish force-sensitive tasks on top of a policy only capable of replicating positions from learned demonstrations with no task reward.

III. DYNAMIC IMITATION VIA RESIDUAL REINFORCEMENT LEARNING

Utilizing a stiff controller for position controlled policies is often sufficient in many tasks given accurate pose tracking at collection time [1]. However, in instances where the position information does not include actions that may induce the desired amount of contact for dynamic imitation such as when loading a scale or scrubbing a table, no amount of stiffness may compensate. Additionally, this information does not exist within the recorded data and must be discovered.

In order to be generally applicable, a learned controller must be usable with any trained position-controlled policy. We train a standard flow matching policy [45], described in Section III-B, with data collected with an adapted version of ALOHA Lightning, found in Section III-A. Given the accuracy of the positions recorded by the motor encoders during data collection, the actual target position corrections that must be made in most instances are small. Thus, reinforcement learning through a residual controller is a natural choice. We define our algorithm and rewards in Section III-D. Learning in simulation may provide strong benefits; however as we are focused on replicating real world contact dynamics the sim-to-real gap can be very large [46]. We add stiffness to the controller policy output, which reduces the sensitivity of positional corrections under contact. Our method, SPAR, as seen in Figure 1 proposes to learn both residual corrective actions and stiffness with real world interaction.

A. Hardware & Data Collection

In our real experiments, we utilize an adapted version of the ALOHA Lightning setup, as a single lightweight backdriveable arm directly grabbed by an operator under gravity and friction compensation. Two Paxini tactile sensors are mounted on the end effector as fingers in a similar fashion to UMI-FT [15], [47]. We only utilize the resultant force stream from both, sending three axes of translation force each at 60hz. We also apply a causal second-order Butterworth filter at 10hz to smooth force readings at both collection and inference time. Beyond this, no analytical adjustments such as a translation to end effector force/torque [15] are made.

In total, for each frame of recorded data we collect measured joint positions $\mathbf{q}_t \in \mathbb{R}^7$ and velocities $\dot{\mathbf{q}}_t \in \mathbb{R}^7$, and a six-axis contact force $\mathbf{f}_{ext,t} \in \mathbb{R}^6$.

The robot is controlled by a joint stiffness controller at 100hz:

$$\boldsymbol{\tau}_t = \mathbf{K}_p \left(\mathbf{q}_t^{\text{cmd}} - \mathbf{q}_t\right) - \mathbf{K}_d \, \dot{\mathbf{q}}_t + c(\mathbf{q}, \dot{\mathbf{q}}), \qquad (1)$$

with the desired joint velocity fixed at zero. $c(\mathbf{q}, \dot{\mathbf{q}})$ is a combined term representing gravity and friction compensation.

The force acting on the end effector of the robot can be characterized by

$$\mathbf{M}\ddot{\mathbf{x}} = \mathbf{f}_{int} + \mathbf{f}_{ext} \qquad (2)$$

where $\mathbf{f}_{int}$, the internal force, is determined by $\boldsymbol{\tau}$, the commanded joint torques. By treating observed joint positions as next step actions ($\mathbf{q}_t^{\text{cmd}} = \mathbf{q}_{t+1}$), the quality of our base policies therefore depends heavily on the choice of $\mathbf{K}_p$ and $\mathbf{K}_d$. A useful heuristic is, if direct action replay with a particular set of gains is successful in a task, the trained policy would too likely be successful.

In our analytical baseline comparisons, we use two separate admittance controller implementations in cartesian end effector space. In simulation, we can directly realize Cartesian impedance because state estimation and force/torque actuation are synchronized and high-bandwidth. On our low-cost hardware platform, low-level motor control is not sufficiently transparent or synchronized for robust Cartesian impedance control. We therefore use an outer-loop admittance controller that converts force error into Cartesian position corrections, which are then converted through inverse kinematics

into joint positions executed by a stiff joint controller. Further, as we mount our force sensors as parallel fingers, they do not return tool-tip cartesian force and thus we can only utilize the pointing direction in our admittance corrections; conversely, in our method, we can learn directly from the measured forces despite not being grounded in the tool tip cartesian frame.

In simulation, to simulate kinesthetic teaching, we use teleoperation for data collection but only utilize position information in the training of the base policies. We hold the action-trained policy as an oracle baseline for comparison.

B. Base Policy

The base policy π_ψ^{base} is a flow matching policy [45] trained with behavior cloning on the demonstrations and frozen thereafter. Conditioned on an observation $\mathbf{o}_t$ of images, proprioception, and force, it predicts a chunk of H future timesteps, jointly containing both a position reference and a contact force:

$$\left(\hat{\mathbf{q}}_{t:t+H},\ \hat{\mathbf{f}}_{t:t+H}\right) \sim \pi_\psi^{\text{base}}(\cdot \mid \mathbf{o}_t)\,. \tag{3}$$

On physical hardware, we use joint positions but in simulation we use end effector cartesian positions. Chunks are predicted asynchronously and executed at the control rate with training time real-time chunking (RTC) [48].

The objective is therefore to map the predicted $\mathbf{f}_{ext}$ implicitly to the requisite joint torques $\boldsymbol{\tau}$ through a residual controller, by adjusting $\mathbf{K}_p$ and applying a delta correction $\Delta\hat{\mathbf{q}}$.

C. Residual Policy

Following [41], [49], we freeze π_ψ^{base} and learn a lightweight residual policy π_θ^{res} that corrects it on hardware. Freezing π_ψ^{base} absorbs the base policy into the environment, as the sampling and chunk scheduling become part of the transition kernel, so the residual faces a stationary MDP $\langle \mathcal{S}^{\text{res}}, \mathcal{A}^{\text{res}}, T, r, \gamma \rangle$ whose actions are bounded corrections. Accordingly, the observation $\mathbf{o}_t^{\text{res}}$ holds a short window of recent position and force tracking errors and measured velocities, π_ψ^{base}'s upcoming references, and the currently applied stiffness; r and the termination set are given in Section III-D.

The residual policy π_θ^{res}, $\mathcal{A}^{\text{res}}$, outputs a normalized joint offset and a stiffness scale,

$$(\Delta\mathbf{q}_t,\ \kappa_t) \sim \pi_\theta^{\text{res}}(\cdot \mid \mathbf{o}_t^{\text{res}})\,, \tag{4}$$

from which the command and the applied gains are formed:

$$\mathbf{q}_t^{\text{cmd}} = \hat{\mathbf{q}}_t + \alpha\,\Delta\mathbf{q}_t, \qquad \mathbf{K}_p = \kappa_t\,\bar{\mathbf{K}}_p. \tag{5}$$

The scale α bounds each joint offset to a fraction of that joint's demonstrated range, and $\kappa_t \in [\kappa_{\min}, 1]$ scales the nominal gains, so the policy may soften the arm but never stiffen it beyond nominal. Both bounds are properties of $\mathcal{A}^{\text{res}}$, so no residual the policy can emit leaves the demonstrated range. Its origin, $\Delta\mathbf{q}_t = \mathbf{0}$ and $\kappa_t = 1$, recovers the base policy exactly.

D. Reward and Learning

Let $\mathbf{e}_t^q = \mathbf{q}_t - \hat{\mathbf{q}}_t$ and $\mathbf{e}_t^f = \mathbf{f}_{ext,t} - \hat{\mathbf{f}}_{ext,t}$ be the position and force tracking errors against the base policy's outputs. The reward at each policy step is

$$r_t = e^{-\|\mathbf{e}_t^q\|/\sigma_q} + \lambda_f \left(e^{-\|\mathbf{e}_t^f\|/\sigma_f} - e^{-\|\hat{\mathbf{f}}_t\|/\sigma_f}\right), \tag{6}$$

where σ_q and σ_f set each term's error scale and λ_f weights force against kinematic fidelity. We choose sigma q and sigma f based on observed maximum deltas when running the base policy. The subtracted term helps avoid bias in the reward from free force-based reward when both predicted and measured forces are zero.

Notice that Eq. (6) carries no task-success term. π_ψ^{base} carries sufficient information for solving the task, and as such reinforcement learning supplies only the force behavior that direct imitation from π_ψ^{base} cannot express.

We optimize π_θ^{res} with an off-policy maximum-entropy actor-critic [50] over a randomized ensemble of critics [51], which supports the high update-to-data ratio required when every transition is collected on physical hardware. The critic Q_ϕ uses the standard bootstrapped target

$$y_t = r_t + \gamma\,(1 - d_t)\,Q_{\phi'}(\mathbf{s}_{t+1},\ \pi_\theta^{\text{res}}(\mathbf{s}_{t+1}))\,, \tag{7}$$

over transitions $(\mathbf{s}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1}, d_t)$ from a replay buffer $\mathcal{D}$.

We treat the task as continuing rather than episodic. Rollouts end at a fixed step budget for operational reasons, and that boundary is recorded as a truncation: Eq. (7) still bootstraps from the final measured state, so the discount factor rather than the rollout length sets the effective horizon.

E. Contact Gating

Given data efficiency concerns in the real world, we also utilize a contact gating mechanism where only chunks predicted by the base policy containing predicted force are eligible for the transition buffer and in which residuals are executable at inference time. This increases the sample efficiency of the learning process and is conceptually similar to ACP [14], which operates at high stiffness and directly lowers stiffness in contact.

IV. Experiments

A. Tasks

In our experiments, we aim to determine how effectively SPAR recovers actions for dynamic imitation, starting with simple tasks such as directly pressing a scale and moving to more complex motions. All chosen tasks are near impossible for position-controlled policies, where

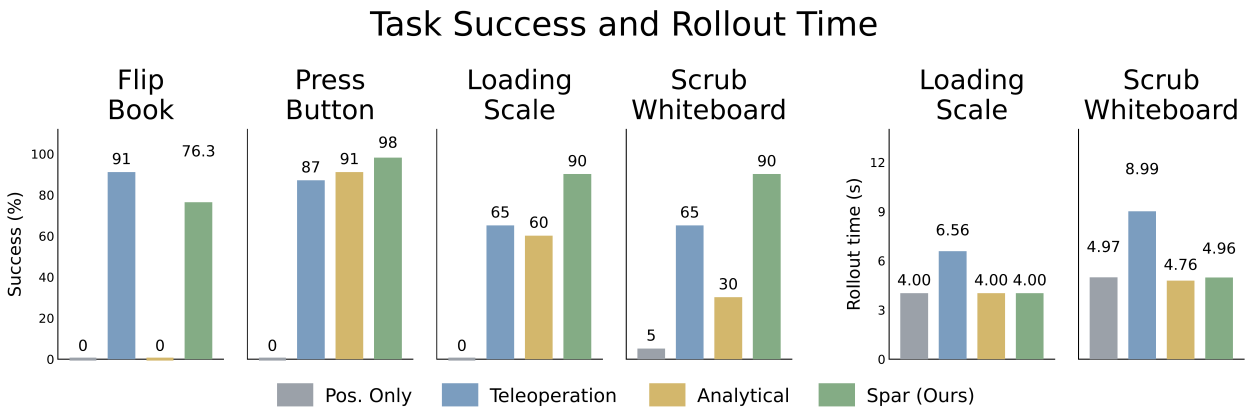


Fig. 2: Task success rate and rollout speed across all four tasks and baselines.

even direct replay of recorded position trajectories fails. In particular, ***Flip Burger*** and ***Scrub Whiteboard*** are highly dynamic tasks with high-frequency contact lasting less than a second, requiring rapid adjustment and precise force prediction. We investigate whether our residual policy, which infers at 100 Hz with a 200 ms observation window, can react to both incoming predictions and changing force sensation. Each task's base position policy is trained on 100 demos. All real tasks are evaluated out of 20, and all sim tasks averaged on three training seeds out of 100. All sim tasks are state-based, and all real tasks are image based position policies.

Flip Burger. In this task setting, we use a 2D simulation based on Push-T [52] with an additional artificial gravity force on a "burger" and collect fast, forceful demonstrations where the demonstrator must push the target of the agent through the burger to exert force, sandwiching it between the agent and the wall, before making a sharp, fast, upwards movement to flip the burger 180 degrees in the air. A success counts as a single flip. We randomize the burger's size and weight, as well as the agent's starting position.

Press Button. In this setting, we use a 3D simulation of a UR5e with Cartesian position control. The objective is to press down on a red button with a sufficient force of 300 N/m and register a task success if the force on the button stays between 300 and 500 N/m for over a second. We randomize the button position.

Loading Scale. In this real-world task analog to ***Press Button***, we press down on a scale with the end effector to register 2 kg of weight, and then flick the tip away. A success in this task includes the number on the scale registering only between 2-2.3 kg, and the flick cleanly disconnecting from the scale and moving away without dragging it. In this task, we use one wrist camera.

Scrub Whiteboard. In this real-world task, we pick up a whiteboard eraser and vigorously scrub a large drawn mark until it is gone. This collection technique contrasts with similar tasks in works such as [15], where wiping a whiteboard is done slowly and in one direction. Success involves erasing the entire mark from the board and returning the eraser to its original place. In this task, we use one wrist and one overhead camera.

B. Comparisons

We register baseline comparisons against ***Analytical***, ***Teleoperation***, and ***Position Only***.

Analytical. We run inference using an analytical approach inspired by a similar design in UMI-FT [15]. We describe our implementation of Cartesian admittance control for this baseline in Section III-A. Using predicted forces and live force readings, we calculate virtual targets live based on calibrated k_{min} and k_{max} to attempt to complete the tasks.

Teleoperation. We compare against policies trained by teleoperating the same arm setup. In sim, this is an oracle that discards action information for our base policies. In the real world, we separately collect teleoperation data with a leader-follower setup on the same device.

Position Only. We compare directly to the base position-tracking policy without online interaction or analytical methods.

C. Online Interaction Improves Tracking

Across all tasks, as shown in Figures 5 and 6, our method reduces force error compared to the position-only policy while maintaining low position error. Figure 3 illustrates the gap between predicted and measured contact forces for the position-only policy. For ***Scrub Whiteboard***, Figure 4 compares representative force trajectories across methods: our residual policy recovers the large contact-force peaks that the position-only and analytical policies fail to reproduce in these rollouts. Finally, this is all done in less than twenty minutes of wall-clock time, being practical while also efficient. The ***Load Scale*** task finishes training at 50k transitions while ***Scrub Whiteboard*** finishes at 100k, both at 100 Hz.

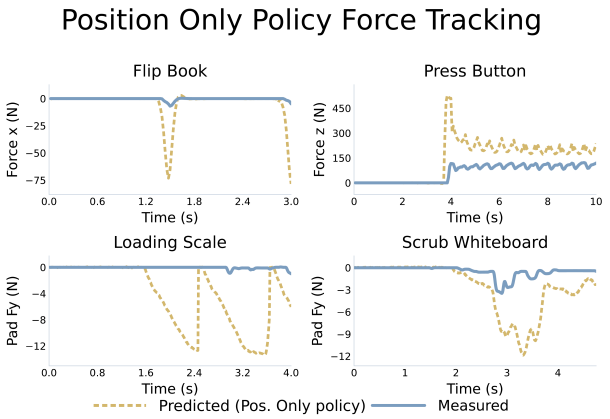


Fig. 3: Across all our tasks, position-only policies fail completely to make necessary contact.

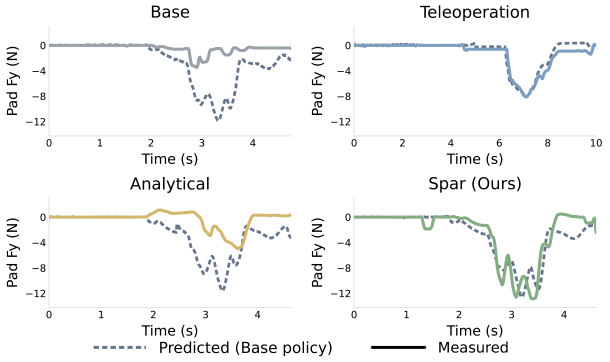


Fig. 4: Scrub Whiteboard comparison over all baselines and our method. SPAR recovers high-frequency components.

D. Speed and Stiffness

In simulation, teleoperation comparisons serve as an oracle, as they possess the action data denied to position-only baselines. However, in real-world experiments, teleoperated policies were up to 2x slower and with poorer force control, resulting in slightly worse policies than our recovered actions collected with direct force feedback. Furthermore, in all tasks, despite not having a reward for low stiffness, SPAR learns to reduce its stiffness during moments of hard contact to more easily discover the appropriate residual actions. This can be seen in Figure 7.

E. Analytical Methods Struggle on High-Frequency Interactions

The analytical method struggles with high frequency oscillations that are present in tasks such as ***Scrub Whiteboard***, and inaccurate pose predictions generated by the base policy in ***Load Scale***, leading to low success rate in Figure 2. In ***Scrub Whiteboard***, the analytical method removes only the middle part of the drawn stars,

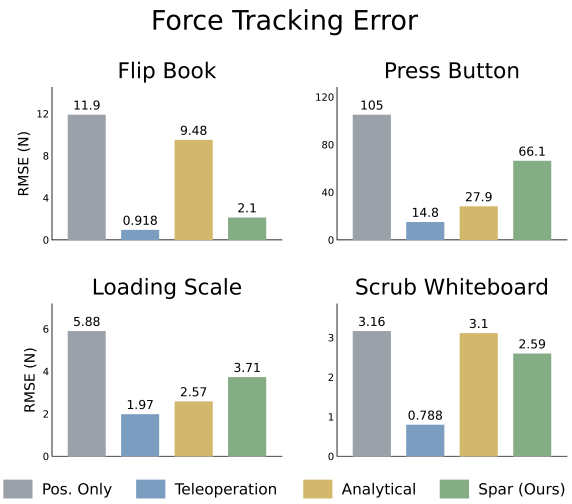


Fig. 5: Normalized force RMSE. Our method routinely closes the gap with force predictions from the base policy.

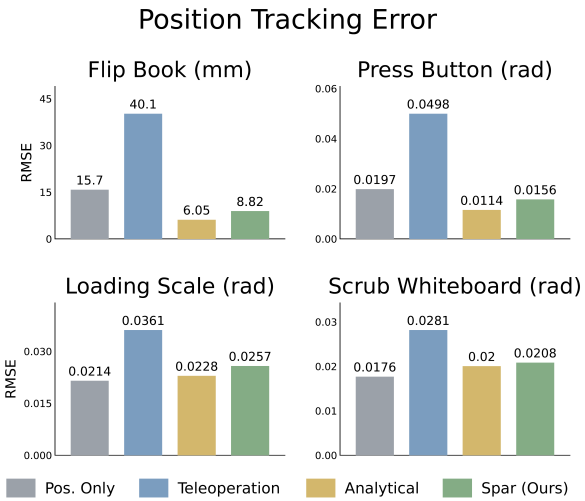


Fig. 6: Our method maintains low position error relative to the predicted positions of a position-only base policy. Teleoperation policies are trained to predict actions, so large position-wise deviations are not an indicator of failure.

failing to adapt its force to fully scrub the drawing as seen in Figure 8. In ***Load Scale***, the analytical method fails on noisy pose predictions, when the fingers contact the scale at an angle. This leads the admittance controller to press too hard on the scale in order to reduce the force error, as contact on a corner of the sensor reads less force than moving fully downwards. In contrast, SPAR learns residual position corrections that lead to more accurate scale loading.

Our dynamic task suite also exposes a limitation of prior methods that restrict compliance adaptation to a single direction [14]. In the ***Flip Burger*** task, the analytical method's unidirectional compliance adaptation

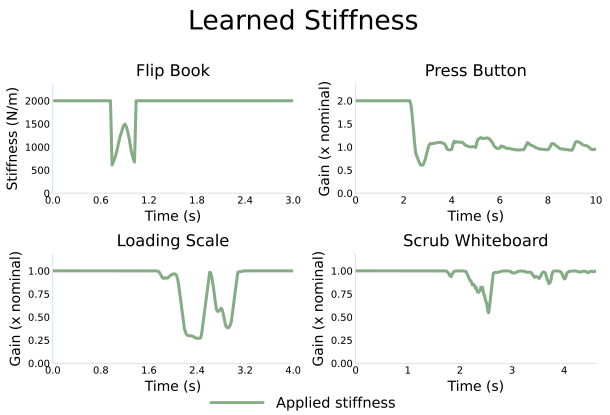


Fig. 7: Stiffness trajectories produced during task execution. SPAR learns to reduce stiffness without direct reward.

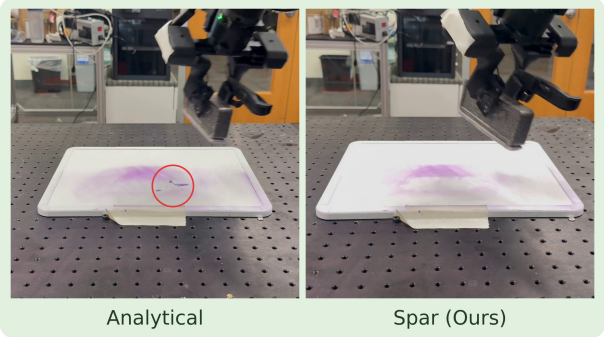


Fig. 8: Analytical methods fail to erase large marks completely, wheras our method fully erases.

resulted in zero success as seen in Figure 2, while SPAR learned to recover the appropriate stiffnesses and actions to reach near the oracle success. Indeed, in a post-hoc experiment we found that simultaneously lowering the stiffness in the perpendicular direction to contact resulted in the analytical method matching the success rate of our method.

Figure 8 compares the whiteboard and robot at the same elapsed time for ***Analytical*** and ***Ours***.

V. Conclusion and Limitations

In conclusion, we present SPAR, a method capable of adapting position-only policies to efficiently recover action data for forceful, high-frequency manipulation. SPAR gives policies the ability to recover complex behaviors, imitating both the kinematics and *dynamics* via a residual policy trained with simple dense reward RL. We find SPAR learns residual positions and stiffnesses in less than 20 minutes of wall-clock time across a wide variety of embodiments and action spaces, and show it outperforms existing analytical methods for recovering action information, while enabling far greater speed than teleoperation. In the future, we aim to investigate more complex force prediction models and further embodiments.

Although direct force prediction performs reliably across the motions and contact conditions we evaluate, our method remains limited by the accuracy of the base policy's force predictions. We apply a causal low-pass filter to suppress high-frequency components and simplify prediction, but this may also remove transient force signals relevant to task execution.

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